

Association between genetically predicted leisure and social activities and cardiovascular disease and other health outcomes

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Participation in leisure and social activities (LSA) is associated with better health outcomes and lower mortality^{1–3}. Previous observational studies demonstrated a relationship between engagement in LSA and both mental and physical health^{4,5}. Although several studies⁶ examined the association between LSA and health outcomes, including cardiovascular disease, their possible causal relationship has not been studied. In this study, we investigated the causal relationship between LSA and various health outcomes, including cardiovascular disease, using data from genome-wide association study summary statistics (sample size = 63,926–1,557,411) and a Mendelian randomization approach. Genetically predicted LSA were associated with a reduced risk of several health outcomes, including coronary artery disease and coronary atherosclerosis. Mediation analysis indicated that these effects were partly mediated by modifiable risk factors, such as body mass index, smoking and lipid levels. These findings highlight the importance of LSA in disease prevention and health promotion.

Participation in leisure and social activities (LSA), such as attending sports clubs, religious groups and adult education classes, is associated with better health outcomes and lower mortality^{1–3}. Previous observational studies demonstrated a relationship between engagement in LSA and both mental and physical health^{4,5}. Although several studies⁶ examined the association between LSA and health outcomes, including cardiovascular disease, their possible causal relationship has not been studied. Reverse causation and residual confounding in observational studies can make validating causal inferences challenging. In addition, the pathway between LSA and health outcomes has not been investigated comprehensively.

The effects of different levels of physical activity on health and longevity have been studied, with research showing that approximately

60–75 min of moderate-intensity activity per day can reduce the risk of death associated with high sedentary time⁷. Another genetic study found that reduced leisure screen time (LST) and higher physical activity levels were linked to lower risk factors mediated by body mass index (BMI)⁸. Similarly, LSA are expected to promote health, although their mechanisms differ from physical activity. Although physical activity mainly benefits health through biological processes (for example, improved cardiovascular health), LSA influence health through psychological (stress reduction and self-esteem), social (strengthening relationships) and behavioral (habit formation) mechanisms⁶. This broader impact positions LSA within a complex adaptive system that influences health through the body and through thoughts, emotions and social interactions.

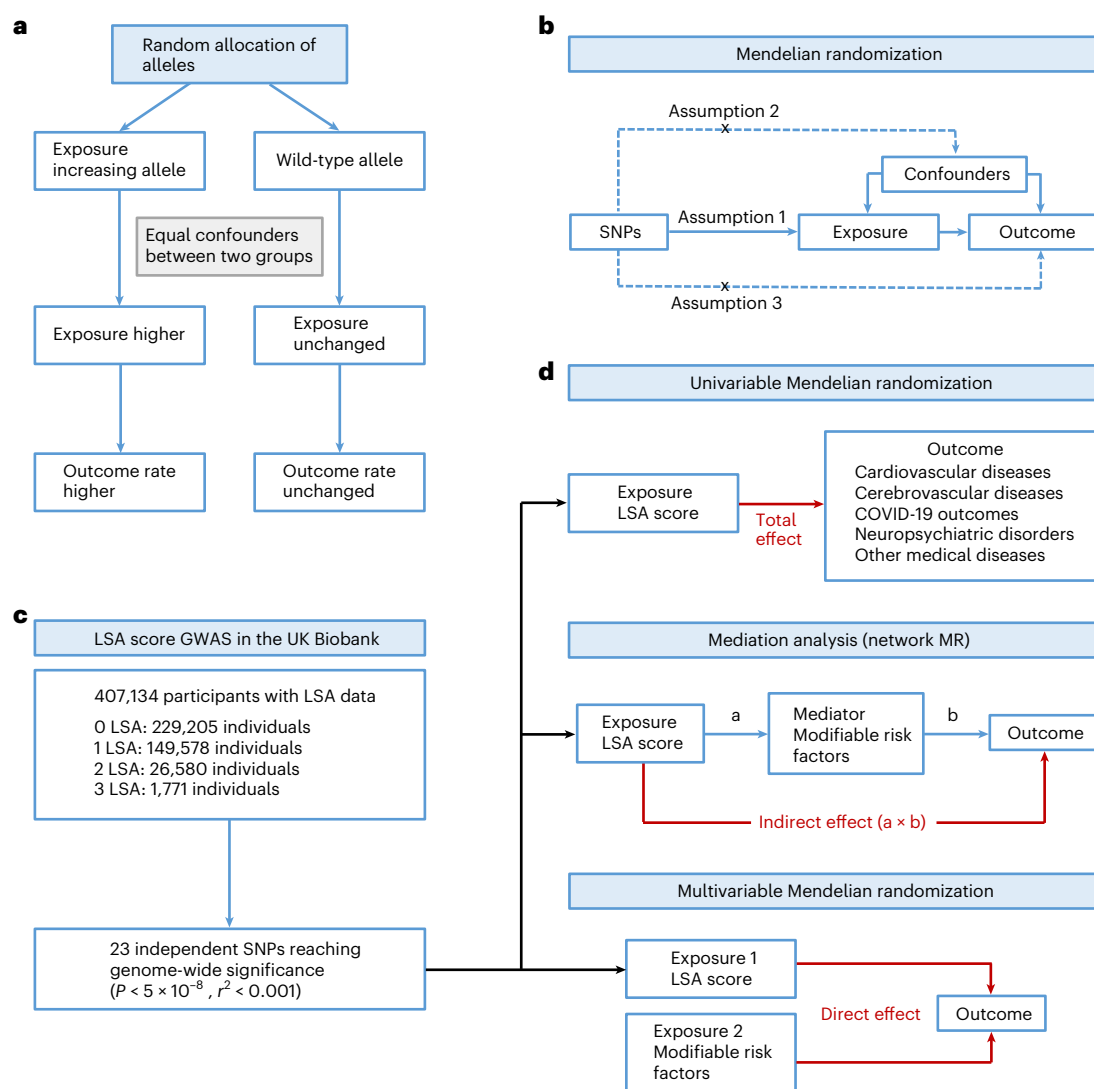


Fig. 1 | Schematic of study design and flow chart. a, MR study design. **b**, Two-sample MR study design. Each GWAS for exposure and outcome was obtained from independent cohorts. **c**, Selection of genetic variants from LSA GWASs as the instrumental variables for use in MR analysis. **d**, Overview of MR methods. r^2 , threshold for clumping.

As genetic variants are assigned randomly during meiosis before birth and are less likely to be confounded with environmental factors, they are useful for inferring potential causal relationships (Fig. 1)⁹. Mendelian randomization (MR) is an approach that can assess the evidence for causality by using genetic variants as instrumental variables (IVs) for an exposure of interest (Fig. 1)^{10,11}. As summary-level results from large-scale genome-wide association studies (GWASs) have become publicly available, the MR method has effectively been applied to a range of behavioral and biological traits^{12,13}.

Since the outbreak of coronavirus disease 2019 (COVID-19), lockdowns and social distancing have been instituted worldwide to impede its spread. Although such measures have been effective in protecting the vulnerable and preventing overstretching of health infrastructure, they curtailed LSA substantially¹⁴. Several studies suggested that such lifestyle changes in the course of the COVID-19 pandemic negatively affected health outcomes associated with both mental and physical health^{6,15}. In such a context, examining the evidence of the potential causal effects of LSA on health outcomes is crucial.

The aim of the present study was to investigate a potential causal association between LSA and various health outcomes, including cardiovascular disease, using MR analysis. To this end, we first conducted

a GWAS to identify genetic variants for LSA as IVs to be used in the MR analysis. We further performed univariable and multivariable two-sample MR and mediation analyses to examine whether engagement in LSA influences health outcomes and whether the effects are mediated by or are independent of other relevant risk factors.

Results

Among the 487,409 individuals in the UK Biobank, a GWAS was conducted for 407,134 individuals of European ancestry, after excluding those without LSA data and after applying sample quality control (QC) for the genotype data. Approximately 54.1% of the study population from the UK Biobank were women, and the mean age at enrollment was 57 years (range, 49–65 years). The demographic characteristics of the participants according to their LSA scores are presented in Supplementary Table 1.

The quantile–quantile plot of the GWAS results revealed genomic inflation ($\lambda = 1.30$), and linkage disequilibrium score regression (LDSC) demonstrated that the inflation was due to polygenicity (LDSC intercept = 1.03, s.e. = 0.009) rather than being derived from population stratification or cryptic relatedness. The 23 genetic variants for LSA (minor allele frequency (MAF) > 3%) presented moderate effect sizes

Table 1 | Description of GWAS summary statistics based on European ancestry for each phenotype

Category	Variable	First author	Consortium	Sample size (cases/controls)
Cardiovascular disease	Coronary artery disease	Nikpay ⁵⁸	CARDIoGRAMplusC4D	141,217 (42,096/99,121)
	Coronary atherosclerosis	–	FinnGen ⁵⁹ (Round 10)	394,668 (51,589/343,079)
	Cardiomyopathy	–	FinnGen ⁵⁹ (Round 10)	318,492 (6,338/312,154)
Cerebrovascular disease	Any type of stroke	Malik ⁶⁰	MEGASTROKE	446,696 (40,585/406,111)
	Any type of ischemic stroke	Malik ⁶⁰	MEGASTROKE	440,328 (34,217/406,111)
	Large-artery atherosclerotic stroke	Malik ⁶⁰	MEGASTROKE	410,484 (4,373/406,111)
	Cardioembolic stroke	Malik ⁶⁰	MEGASTROKE	413,304 (7,193/406,111)
	Stroke caused by small-vessel disease	Malik ⁶⁰	MEGASTROKE	411,497 (5,386/406,111)
COVID-19 outcome	COVID-19 infection ^{a,b} (C2)	–	COVID-19 HGI ⁶¹ (Round 5)	1,348,701 (32,494/1,316,207)
	COVID-19-associated hospitalization ^{a,b} (B2)	–	COVID-19 HGI ⁶¹ (Round 5)	1,557,411 (8,316/1,549,095)
	COVID-19 death or respiratory failure ^{a,b} (A2)	–	COVID-19 HGI ⁶¹ (Round 5)	1,059,456 (4,792/1,054,664)
Neuropsychiatric disorder	MDD ^{a,b}	Wray ⁶²	PGC	173,005 (59,851/113,154)
	Bipolar disorder	Mullins ⁶³	PGC	413,466 (41,917/371,549)
	Schizophrenia	Trubetskoy ⁶⁴	PGC	130,644 (53,386/77,258)
	Alzheimer's disease	Kunkle ⁶⁵	–	63,926 (21,982/ 41,944)
	Parkinson's disease	–	FinnGen ⁵⁹ (Round 10)	412,181 (4,681/407,500)
Other medical disease	Asthma	–	FinnGen ⁵⁹ (Round 10)	386,387 (48,084/338,303)
	Breast cancer	Michailidou ⁶⁶	–	139,274 (76,192/63,082)
	LC	McKay ⁶⁷	–	85,716 (29,266/56,450)
	RA	–	FinnGen ⁵⁹ (Round 10)	276,465 (13,621/262,844)
Modifiable risk factor	Physical activity (overall activity) ^c	Doherty ¹⁸	UK Biobank	91,105
	Type 2 diabetes	–	FinnGen ⁵⁹ (Round 10)	400,197 (65,085/335,112)
	Systolic blood pressure	Ehret ⁶⁸	ICBP	69,395
	BMI	Locke ⁶⁹	–	322,154
	LDL cholesterol	Graham ⁷⁰	GLGC	842,660
	HDL cholesterol	Graham ⁷⁰	GLGC	888,227
	TGs	Graham ⁷⁰	GLGC	864,240
	Cigarettes per day ^{a,b}	Saunders ⁷¹	–	183,196
	Drinks per week ^{a,b}	Saunders ⁷¹	–	304,322

CARDIoGRAMplusC4D, coronary artery disease genome-wide replication and meta-analysis plus the coronary artery disease genetics; COVID-19 HGI, coronavirus disease 2019 caused by SARS-CoV-2 Host Genetics Initiative (release 5); GLGC, Global Lipids Genetics Consortium; ICBP, International Consortium for Blood Pressure; PGC, Psychiatric Genomics Consortium.

^a23andMe data are excluded from the discovery GWAS. ^bUK Biobank data are excluded from the discovery GWAS. ^cTo remove the sample overlap between LSA data and physical activity data ($n=86,872$), we performed GWAS analysis for LSA with the non-overlapping individuals ($n=320,262$). These GWASs were all of European ancestry and of both males and females.

ranging from -0.06 to 0.04 . The single-nucleotide polymorphism (SNP) heritability of LSA was estimated to be 5.7% (95% confidence interval (CI): $5.2-6.2$).

We performed the GWAS using a proportional odds logistic mixed model (POLMM)¹⁶ owing to the ordinal nature of the LSA score, which ranges from 0 to 3. This analysis identified 23 independent genomic loci associated with LSA at a genome-wide significance threshold ($P < 5 \times 10^{-8}$, $r^2 < 0.001$) and considered the most significant SNPs in each locus as lead SNPs (Fig. 1, Extended Data Fig. 1 and Supplementary Table 2). Among the significant loci in GWAS for LSA, only one SNP (rs4470910) was near a significant locus in the previous GWAS for physical activity⁸. It suggests that LSA is a trait distinct from physical activity.

According to the GWAS catalog manual curation, 10 of the 23 loci were associated with a variety of human traits, including socioeconomic indicators (education and income), health behaviors (smoking cessation, physical activity and sleep), physical characteristics (height, weight and body fat percentage), mental characteristics (educational achievement, math ability, brain volume, depression and

mood disorder) and diseases and biomarkers (cardiovascular disease, BMI, high-density lipoprotein (HDL) cholesterol, diabetes and hemoglobin measurement). Notably, when compared to physical activity, LSA exhibited associations with a broader range of multidimensional human traits, suggesting complex underlying mechanisms involving interactions within the LSA framework⁶. Detailed results are presented in Supplementary Table 3.

Cell-type-specific heritability was enriched in the central nervous system (CNS), including the cortex region, limbic system, hippocampus and basal ganglia, at false discovery rate (FDR) $< 5\%$ (Extended Data Fig. 2 and Supplementary Tables 4 and 5). These enrichments in the CNS region were replicated in multiple tissues of the chromatin dataset. Among three cell types in the CNS region (astrocyte, neuron and oligodendrocyte), neurons were markedly enriched for heritability (Supplementary Table 6).

In the genetic correlation analysis between LSA and 29 traits, 23 traits (15 health outcomes and eight modifiable risk factors) were significantly related to LSA at FDR $< 5\%$ (Extended Data Fig. 3 and

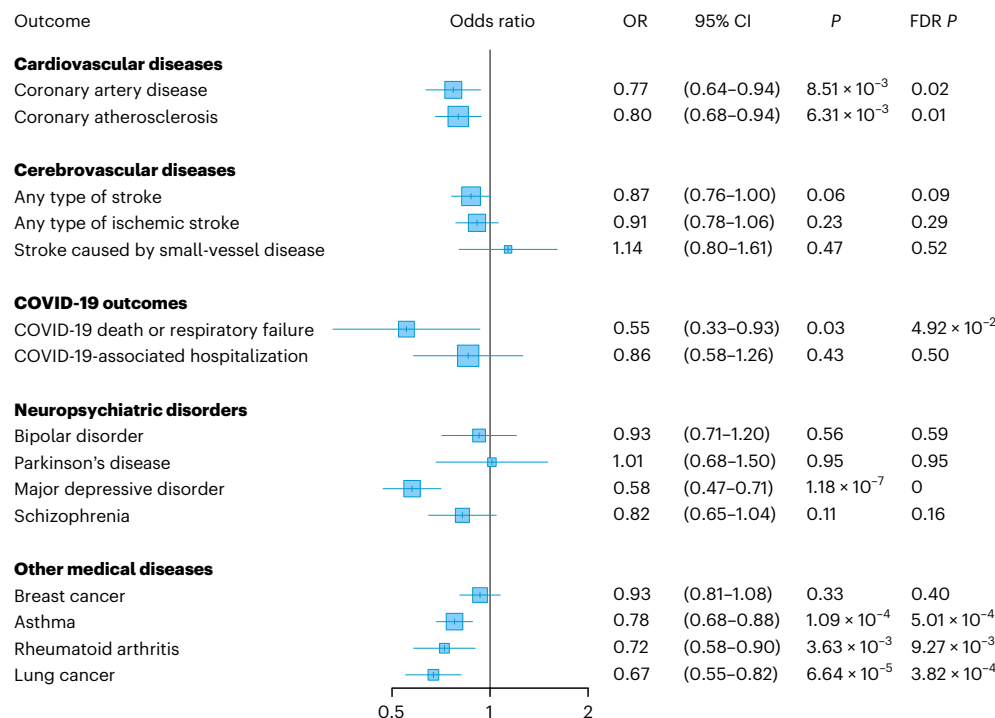


Fig. 2 | Forest plot for the univariable MR showing effects of genetically predicted LSA on 15 health outcomes. Total variants used in the analysis was $n_{\text{variant}} > 10$. Data are presented as mean $\pm$ s.d. Statistical analyses were conducted using two-sided tests. Error bars represent the 95% CIs of the effects.

Supplementary Table 7). These 23 traits were used either as an outcome or as a mediator in the MR analysis. Among the significant modifiable risk factors and health outcomes, cigarettes per day ($r_g = -0.33$, $P = 3.09 \times 10^{-16}$) and lung cancer (LC) ($r_g = -0.39$, $P = 1.44 \times 10^{-13}$) revealed a genetic correlation greater than 0.3. We found a low genetic correlation between LSA and physical activity ($r_g = 0.23$, $P = 3.02 \times 10^{-16}$). This indicates that LSA may have genetic components distinct from physical activities and may include aspects of social interaction. In addition, we examined the genetic correlations between our LSA score and the leisure-time-specific physical activity traits reported by Wang et al.⁸. The correlation between LSA and moderate-to-vigorous intensity physical activity (MVPA) during leisure time was $r_g = 0.79$ ($P \approx 0$). In contrast, LSA was negatively associated with LST, which typically includes time spent watching TV, playing video games or using computers ($r_g = -0.53$, $P = 1.33 \times 10^{-119}$).

In the primary analysis, we performed a univariable MR analysis to estimate the association between genetically predicted LSA and health outcomes (Table 1)¹⁷. We performed a GWAS analysis for LSA with non-overlapping individuals ($n = 320,262$) in addition to the primary GWAS to remove the sample overlap between LSA data and physical activity data ($n = 86,872$)¹⁸ (Extended Data Fig. 4 and Supplementary Table 8). Supplementary Table 9 shows all associations of 'genetic variant–exposure' and 'genetic variant–outcome', with harmonized effects and alleles. The odds ratio (OR) from the univariable MR analysis refers to the change in odds of an outcome for each incremental change in LSA score. Using the inverse variance weighted (IVW) method, we found evidence that a lower LSA score was significantly associated with the risk of seven health outcomes (FDR-corrected $P < 0.05$; Fig. 2 and Supplementary Table 10): coronary artery disease (CAD) (OR = 0.77, 95% CI: 0.64–0.94, $P = 8.51 \times 10^{-3}$), coronary atherosclerosis (CA) (OR = 0.80, 95% CI: 0.68–0.94, $P = 6.31 \times 10^{-3}$), COVID-19 death or respiratory failure (severe COVID-19 outcome) (OR = 0.55, 95% CI: 0.33–0.93, $P = 2.58 \times 10^{-2}$), major depressive disorder (MDD) (OR = 0.58, 95% CI: 0.47–0.71, $P = 1.18 \times 10^{-7}$), asthma (OR = 0.78, 95% CI: 0.68–0.88, $P = 1.09 \times 10^{-4}$), LC (OR = 0.67, 95% CI: 0.55–0.82, $P = 6.64 \times 10^{-5}$) and rheumatoid arthritis (RA) (OR = 0.72, 95% CI: 0.58–0.90, $P = 3.63 \times 10^{-3}$).

The intercept test of MR-Egger regression demonstrated that all these significant univariable MR effects were not influenced by directional horizontal pleiotropy, except for CA (P for the MR-Egger intercept test > 0.05)¹⁹. Moreover, estimates from the weighted median (WM) method, conducted as a sensitivity analysis, were similar to the estimates from the IVW method, except for CA. For the seven outcomes, the leave-one-out sensitivity analyses for the IVs showed that there was no single genetic variant with a substantial effect on the overall estimates of MR (Supplementary Figs. 1–7). The MR Steiger test reliably showed the correct direction of the potential causal relationship between LSA and health outcomes (FDR-corrected $P_{\text{Steiger}} < 0.05$) (Supplementary Table 11).

We assessed the validity of the three core assumptions for MR to confirm the reliability of our analyses. To satisfy the first assumption, we selected SNPs strongly associated with LSA ($P < 5 \times 10^{-8}$). For the second assumption, we confirmed that all GWAS data used in two-sample MR analysis had no genetic relatedness, which may interfere with the random assignment of genetic variants within one population²⁰. The third assumption was fulfilled by confirming the lack of significant association between the instruments and the outcomes ($P > 5 \times 10^{-8}$) and by the MR-Egger regression test, which showed no evidence of bias-generating pleiotropy.

In step 1, genetically predicted LSA were significantly associated with five out of the eight genetically correlated modifiable risk factors using the IVW method at the FDR 5% threshold: triglycerides (TGs) (beta = -0.09 , 95% CI: -0.14 to -0.04 , $P = 6.98 \times 10^{-4}$); BMI (beta = -0.14 , 95% CI: -0.22 to -0.06 , $P = 4.14 \times 10^{-4}$); cigarettes per day (smoking) (beta = -0.13 , 95% CI: -0.20 to -0.06 , $P = 2.24 \times 10^{-4}$); HDL cholesterol (beta = 0.16 , 95% CI: 0.11 – 0.21 , $P = 1.29 \times 10^{-10}$); and low-density lipoprotein (LDL) cholesterol (beta = -0.03 , 95% CI: -0.06 to -0.003 , $P = 2.78 \times 10^{-2}$) (Extended Data Fig. 5). In step 2, we used a bidirectional MR analysis to evaluate if modifiable risk factors that showed significant genetic correlations with LSA had a reverse directional effect on LSA. We found that, in the reverse MR analysis, with type 2 diabetes as the exposure and LSA as the outcome, the result was significant (OR = 0.97, 95% CI: 0.96–0.99, $P = 2.50 \times 10^{-3}$), suggesting a bidirectional

relationship between LSA and type 2 diabetes, on account of which we excluded them from further analysis (Extended Data Fig. 6).

Moreover, we explored whether the five candidate mediators (BMI, TGs, HDL cholesterol, LDL cholesterol and smoking) had a potential causal effect on each of the seven health outcomes that were significant at the FDR 5% threshold in the primary analysis. We used the five candidate mediators as the exposure in step 3 and evaluated the association of 35 pairs (5×7). A total of 14 pairs had significant associations when using the IVW method at the FDR 5% threshold for each outcome (Extended Data Fig. 7). The mediating effects of modifiable risk factors between LSA and health outcomes varied. For instance, the indirect impacts of BMI, HDL cholesterol, LDL cholesterol and TGs were significant for both CAD and CA, whereas those of BMI and smoking were significant for asthma and RA (Supplementary Table 12 and Extended Data Fig. 8). For LC, only one of the factors examined, smoking, accounted for approximately 59% of the total effect. We found no significant indirect effects on severe COVID-19 outcome and MDD. These findings demonstrate that several different pathways exist between LSA and each health outcome, some of which may still need to be identified. With the exception of the LSA–LDL–asthma triad, all indirect effects of LSA via the 13 out of 14 mediator–outcome pairs were significantly protective. The proportions of the mediated effect of LSA via each mediator ranged from 5% to 59% of the total effect.

We applied multivariable MR (MVMR) analysis to the 14 mediator–outcome pairs identified in the mediation analysis. The direct effects of LSA on each outcome remained significant in eight out of 14 mediator–outcome pairs after each mediator was considered (Fig. 3 and Supplementary Tables 13 and 14). We calculated the conditional *F*-statistics for each exposure in the MVMR analyses and observed high *F*-statistic values for the exposures except LSA and LDL cholesterol. The MVMR analysis revealed a direct effect of LSA on CAD, CA and asthma, even after accounting for BMI and LDL cholesterol. Notably, the significance of LSA persisted even when considering BMI or smoking status in relation to RA and LC. This indicates that the association between genetically predicted LSA and several health outcomes was robust, not only through the mediators but also independent of the mediators.

Discussion

In the present study, we examined the potential causal association between LSA and health-related traits, including cardiovascular disease, and their mediating factors, using the GWAS results for LSA from this study and summary statistics obtained from large-scale GWASs of other traits. LSA were genetically correlated with seven modifiable risk factors and 15 health outcomes. In the subsequent MR analyses, we found evidence for the potential causal protective effects of LSA on the various health outcomes.

The 23 LSA-associated genetic variants were associated with various phenotypes in previous findings. The results obtained from expression quantitative trait loci (eQTL) gene mapping, pathway and LDSC analyses demonstrated that the genetic architecture of LSA was mainly associated with the CNS and anti-inflammatory system, and LSA were genetically correlated with various modifiable risk factors and health outcomes.

Our MR findings support the protective effect of LSA on health outcomes, similar to previous studies^{21–23}. Mediation analyses showed that LSA influenced modifiable risk factors, such as BMI, smoking and lipid metabolism (HDL/LDL cholesterol and TGs), which sequentially impacted health outcomes. This aligns with previous epidemiological findings^{24–26}. MVMR analysis confirmed that LSA had a direct protective effect on health. In eight out of 14 LSA mediator–outcome triads, the impact of LSA remained significant even after adjusting for five key mediators. The effects of LSA on LC remained significant after adjusting for smoking, with a notable indirect effect on smoking (59%). The lack of significant effects on severe COVID-19 and MDD suggests that other, yet unknown, mediators may link LSA to health outcomes.

LSA may decrease stress, enhance positive mood and reduce low-grade inflammation²⁷. Therefore, an increase in LSA may help protect against health issues by reducing stress and inflammatory indicators²⁸. A greater involvement in LSA may reduce sedentary behavior and enhance physical activity, improving cardiometabolism^{25,29}. From a behavioral perspective, LSA engagement may lead to healthy habits, better health-related choices and increased health access^{6,22}. For example, as activity levels grow, smoking and BMI tend to decrease^{24,26}. This explanation may be attributed to physical activity. However, genetic correlation results indicate that LSA cannot be solely accounted for by physical activity measured by accelerometers ($r_g = 0.23$) (Supplementary Table 7). In contrast, LSA was negatively associated with LST ($r_g = -0.53$). Although the traits were correlated, only three significant loci (rs11687241, rs7613360 and rs7015700; $r^2 > 0.1$) were common among them, indicating distinct genetic components. The overlap for the LSA score was 13% (three out of 23 significant loci). Overall, LSA are positively associated with physical activity during leisure time and negatively associated with sedentary traits. This negative correlation suggests that higher LSA scores correspond to less time spent on sedentary leisure activities. LST is mainly characterized by sedentary behavior and lacks social interaction, which may explain the negative correlation with LSA. Previous GWAS results demonstrated that loci for LSA were associated with traits across various domains, including social, behavioral, physical and mental traits (Supplementary Table 3). The leisure phenotypes used by Wang et al.⁸ and those in the present study differ mainly in definitions. Wang et al.⁸ focused on leisure time physical (in) activity, whereas we combined three activities (sports clubs, religious groups and adult education classes) into a single LSA score to describe leisure time activities. These items showed high genetic correlations, ranging from 73% to 80% (Supplementary Table 15), suggesting shared genetic effects and justifying their combination into a single LSA score for analysis. We conducted a separate GWAS on individual LSA items and examined their relationships with various health outcomes to evaluate causal effects. In the MR analysis, the IVW method revealed that engagement in religious activities and adult education classes positively affected health outcomes (Supplementary Tables 16 and 17). This suggests that combining physical and social activities might have a more substantial impact on health than participation in a single component, such as physical activity alone.

The neuro-endocrine-immune system may offer insights into the link between LSA and health outcomes. LSA-induced immune activity regulates substances such as C-reactive protein, interleukin-6 and TNF, which affect inflammation³⁰. The hypothalamic–pituitary–adrenal (HPA) axis, which controls these substances via cortisol secretion³¹, may also play a role in LSA. Enrichment of SNPs for CNS tissues and findings from eQTL analysis suggest that both the CNS and immune systems are involved in LSA, supporting this hypothesis. Mediators regulated by the HPA axis may further explain the impact of LSA on health outcomes.

Our results suggest that LSA may protect against certain diseases and improve health. Although most clinical guidelines focus on traditional risk factors³², the role of LSA in prevention has been overlooked compared to physical activity. Clinicians should encourage greater LSA engagement to improve public health, particularly for cardiovascular care. Promoting a healthy lifestyle through LSA can enhance quality of life, and health authorities should prioritize LSA interventions.

Our study does have, however, several limitations. First, the study samples, including UK Biobank participants and other GWASs, were limited to individuals of European ancestry, which may reduce confounding by population stratification but limits the generalizability of our findings to other ethnicities. Second, we calculated the conditional *F*-statistics for MVMR to assess the strength of IVs. For some exposures, particularly LSA, the conditional *F*-statistic did not exceed the robustness threshold (conditional *F*-statistic > 10), suggesting a possible weak instrument for MVMR. This may be due

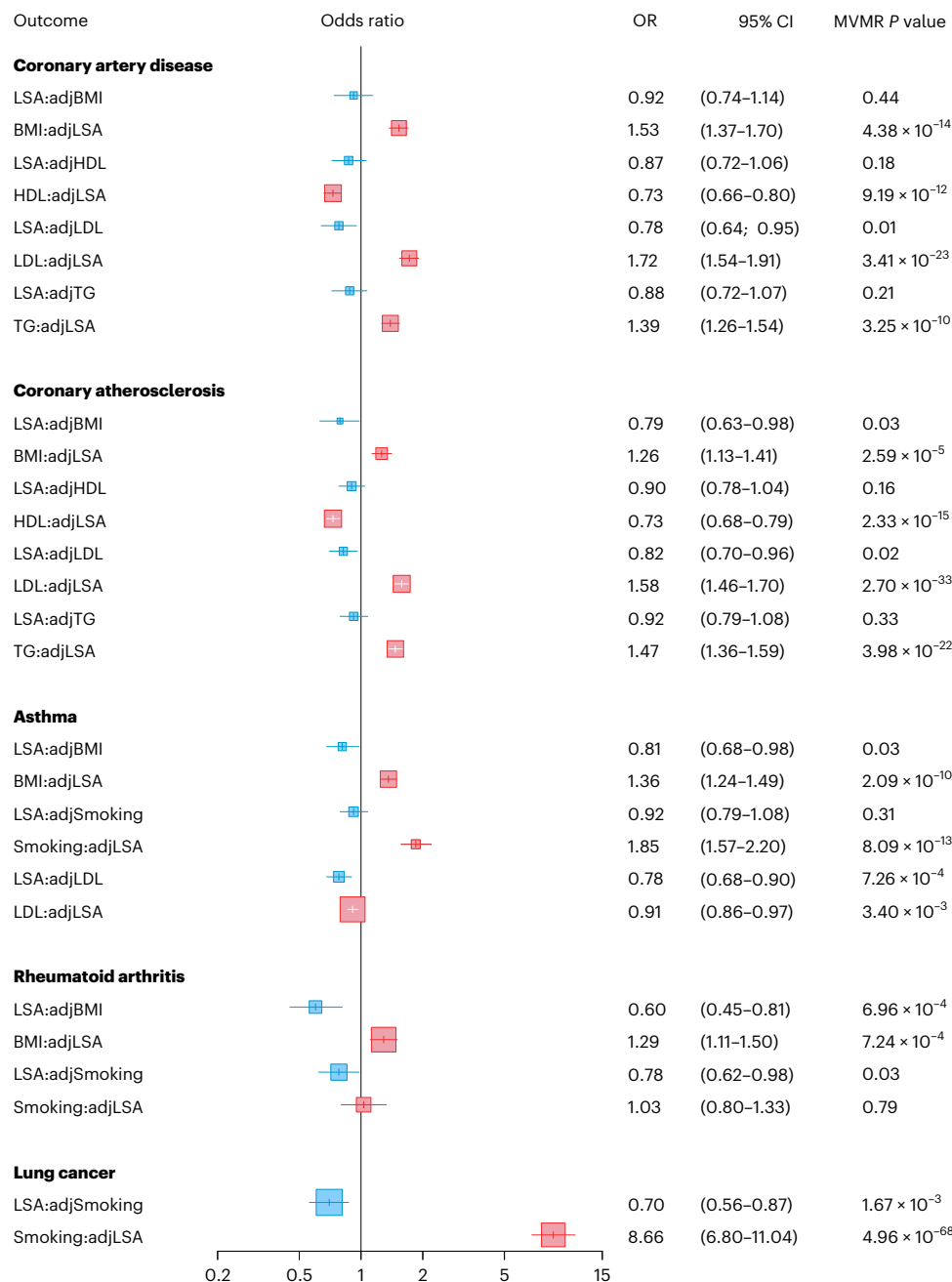


Fig. 3 | Forest plot for the MVMR showing effects of genetically predicted LSA on five health outcomes accounted for by selected mediators. Total variants used in the analysis was $n_{\text{variant}} > 10$. Data are presented as mean $\pm$ s.d. Statistical

analyses were conducted using two-sided tests. Error bars represent the 95% CIs of the effects.

to the close relationship between the two exposures included in the estimation. Therefore, caution is required when interpreting MVMR results. Third, the measurement of LSA based on a questionnaire may not capture all variations or changes in LSA, leading to potential residual bias. The LSA score, calculated from weekly participation, does not account for individual differences over time. Despite these limitations, our approach robustly identifies genetic variants associated with LSA and their health benefits. Future studies should include diverse populations and more comprehensive LSA measurements. Our MR and mediation analyses indicate that LSA influence various health outcomes, with potential causal effects partly mediated by modifiable risk factors. These findings suggest that engagement in LSA can effectively target the prevention of chronic diseases and promote public health.

Methods

This study adheres to all relevant ethical regulations. It was approved by the institutional review board of Seoul National University Bundang Hospital (X-2202-737-902). Each GWAS summary statistic used in this study was previously approved by its respective research ethics committee (Table 1). The UK Biobank study was approved by the National Research Ethics Committee (REC reference 11/NW/0382). All UK Biobank participants provided informed consent. The UK Biobank data were obtained under application number 33002.

Study design, data sources and GWAS

We obtained LSA data from the UK Biobank³³. A total of 407,134 individuals of European ancestry were used as a GWAS cohort for LSA. SNPs that were strongly associated with LSA ($P < 5 \times 10^{-8}$) in the GWAS

were used as IVs in MR for estimating its effect on health outcomes. The overall study design is illustrated in Fig. 1. For the risk factors and outcomes of interest for MR, we obtained GWAS summary statistics from previously published studies: the MR-Base database and the GWAS catalog (Table 1)³⁴. Of more than 1,000 traits with available GWAS results, 29 were chosen based on several criteria for traits to be analyzed. First, we included only common complex diseases/conditions with a sample size exceeding 4,000 to ensure robust estimation of genetic correlation using LDSC software³⁵. Second, summary statistics were primarily derived from individuals of European ancestry, excluding the UK Biobank, to avoid sample overlap with the LSA GWAS. Third, the sample sizes between cases and controls for binary trait GWAS should be relatively balanced. Fourth, we selected summary statistics with more than four independent genome-wide significant loci for use as IVs, allowing the application of various MR methods, including MR pleiotropy residual sum and outlier (MR-PRESSO), which requires more than three variants to detect outliers.

LSA definition

We defined LSA using the UK Biobank data-field 6160 on leisure/social activities ('Which of the following do you attend once a week or more often? [you can select more than one]'). The selected activities were 'sports club or gym', 'religious group' and 'adult education class', with a score of 0–3 based on the number of 'yes' responses. 'Pub or social club' was excluded due to its association with alcohol consumption. Participants with missing data or 'prefer not to answer' were excluded, leaving 407,134 for analysis. For the GWAS analysis, LSA engagement was coded as 'engagement in a specific LSA item' versus 'no engagement in any leisure activity'.

LSA GWAS for genetic instrumental variables

UK Biobank samples were genotyped using Affymetrix UK BiLEVE Axiom and Affymetrix UK Biobank Axiom arrays released in March 2018. Phasing was performed using SHAPEIT3 (ref. 36), and imputation was carried out centrally by the UK Biobank using a combined panel (1000 Genomes Project and UK10K) and IMPUTE2 (ref. 37). The following variant-level QC exclusion metrics were applied to imputed data for GWAS: call rate < 95%, Hardy–Weinberg equilibrium $P < 1 \times 10^{-6}$ and MAF < 1×10^{-4} . Additionally, stringent QC using PLINK 1.90 software was performed. We excluded the variants if call rates were less than 0.95, MAF values were less than 0.005 for related samples or imputation quality scores (INFO) were less than 0.4; genotypes with a posterior call probability less than 0.90 were considered missing, which resulted in the remaining 9,575,249 SNPs that passed the QC criteria. The sample-level QC exclusion metrics applied to the imputed GWAS data were as follows: non-European samples, samples with mismatched sex, putative sex chromosome aneuploidy, no sex information and participants who withdrew from the UK Biobank.

The LSA score, which ranges from 0 to 3, has ordered categories; however, there are no measurable units with uniform intervals between them. Therefore, for this ordinal categorical phenotype, we performed a GWAS using a POLMM¹⁶. POLMM uses a sparse genetic relationship matrix to adjust for relatedness within samples as a random effect and uses a saddle point approximation to control inflated type I error rates owing to the unbalanced distribution in ordinal categorical data. The effect estimates from this GWAS result for LSA were used as the effects of exposure in the subsequent MR analysis.

A genome-wide significance threshold of $P < 5 \times 10^{-8}$ was used to identify LSA-associated variants. The most significant SNPs with the lowest P value in the corresponding genomic loci were selected as the lead SNPs. The maximum distance for linkage disequilibrium (LD) blocks to merge into a genomic locus was 3,000 kb. Genetic data of the European population in the 1000 Genomes Project v3 were used for the LD calculation. ANNOVAR³⁸ implemented in FUMA³⁹ was used to annotate SNPs.

For the independent genomic risk loci, evidence of eQTL and functional annotation was examined using the FUMA platform⁴⁰. Mapping SNPs to genes significantly associated with eQTL were mapped using the Genotype-Tissue Expression (GTEx) database v8 (ref. 41). Biological pathway analysis was performed on results from gene-based analysis using MAGMA implemented in FUMA based on the Gene Ontology Consortium⁴².

GWAS catalog manual curation

Using FUMA, we examined whether the loci identified in our GWAS overlapped with previously reported loci listed in the NHGRI-EBI GWAS Catalog⁴³. Lead variants identified in our study were manually curated using the GWAS catalog. LD was calculated for variants within ± 500 kb, and variants with $r^2 > 0.8$ were defined as LD proxies.

Estimation of SNP heritability and genetic correlation

We used LD score regression analysis with LDSC software to calculate the SNP-based heritability for LSA⁴⁴. We applied LDSC for specifically expressed genes (LDSC-SEG), a method that contrasts the expression level in a selected tissue with that of all other tissues to examine tissues in which the genes associated with LSA were differentially expressed⁴⁵. Before the MR analyses, we estimated the genetic correlations between LSA and 29 traits (20 health outcomes and nine modifiable risk factors; Table 1) using LDSC because correlation assesses the extent of the shared genetic etiology between traits and owing to a causal relationship³⁵. We applied multiple testing corrections in this analysis by using an FDR of 5%.

Primary analysis for MR

MR is an epidemiologic method based on IV analysis and employs genetic variants, mainly SNPs, as IVs associated with a modifiable exposure, which are then used as proxy indicators to infer possible causal relationships with a health outcome¹⁰. Each IV (that is, genetic variant) used in the analysis must satisfy three assumptions for the effect to be estimated consistently: (1) IV is strongly associated with the exposure; (2) IV is not associated with any confounders of the exposure–outcome association; and (3) IV influences risk of the outcome via exposure (vertical pleiotropy), not through other pathways (horizontal pleiotropy)^{17,46}. These key assumptions of IVs are illustrated in Fig. 1b. To assess and enhance the quality of our MR analyses, we adhered to a checklist based on Strengthening the Reporting of Observational Studies in Epidemiology of Mendelian Randomization (STROBE-MR) guidelines⁴⁷ (Supplementary Table 18). This table includes detailed descriptions of the measurement, quality and selection of genetic variants and the validity of the key assumptions underlying MR.

We performed the two-sample univariable MR to estimate the total effect of LSA on health outcomes. For this analysis, we used the IVW method, which provides an estimate of effect under the assumption that all genetic variants are valid as IVs^{18,48}. Therefore, the potential causal effect estimates of each genetic variant are combined in a weighted average of the estimates.

Regarding IVs for the exposure of interest (LSA), we clumped SNPs exhibiting genome-wide significant associations ($P < 5 \times 10^{-8}$) with r^2 values ≤ 0.001 (clumping window size = 1 Mb) using the European population reference from the 1000 Genomes Project v3. We excluded genetic variants that were non-existent in the health outcome GWASs or non-inferable palindromic SNPs. The excluded variants were replaced by LD proxy SNPs ($r^2 < 0.8$) where available. We harmonized the effects (beta coefficients) of genetic variants on LSA and health outcomes for the same allele. We calculated the F -statistics to assess the robustness of the genetic variants as instruments for the exposure GWAS⁴⁹. Additionally, we computed the conditional F -statistic to assess the strength of IVs after adjusting for the association between exposures in the MVMR model, using the MendelianRandomization R package (version 0.9.0)⁵⁰. The MR Steiger test was used to infer the direction of

a possible causal link between LSA and outcome traits⁵¹. We applied the FDR to the univariable MR (FDR threshold of 5%) to account for multiple comparisons.

Sensitivity analysis for MR

The primary assumption regarding the examination of a potential causal relationship using MR is that the IVs have no horizontal pleiotropy. If the IVs violate this assumption, the effect estimate may be biased. Thus, we performed the MR-PRESSO method to detect and remove IVs with potential pleiotropy⁵². We applied the MR-PRESSO global test to the primary MR result to identify possible bias from horizontal pleiotropy after excluding any outliers⁵². MR-PRESSO consists of three parts: a global test, an outlier test and a distortion test. We conducted all three steps, detected outliers and excluded them from subsequent analysis. Additionally, we tested the third IV assumption that genetic instruments affect the outcome exclusively through the exposure of interest using MR-Egger regression. We obtained the intercept of the MR-Egger regression to test for pleiotropic effects. This analysis allowed for directional pleiotropic effects, indicating that some SNPs might influence outcomes through pathways other than exposure. This is a relaxed IV assumption referred to as the InSIDE assumption (INstrument Strength Independent of Direct Effect)⁴⁹. The slope of the MR-Egger regression represents a pleiotropy-corrected effect, and the average pleiotropic effects of the IVs can be estimated using the intercept term, which differs from zero⁴⁹. An intercept value that is not equal to zero indicates evidence of directional pleiotropy.

For a robust inference, we compared the IVW results with the WM method. The WM method assumes that at least half of the total weight of the IVs comes from valid variants⁵³. Given more than 50% of the true effect, this method provides a more reliable estimate because the genetic variation contributing more strongly to the potential causal effect has more weight. We also estimated *F*-statistics to confirm instrument strength, suggesting that the IVs are valid when the *F*-statistic is greater than 10 (ref. 54).

Mediation analysis

Mediation analysis using a two-step MR can be used to assess whether modifiable risk factors mediate the effects of LSA on health outcomes. Following a previous study, we set up four sequential steps of mediation analysis for each pair within the exposure (X)–mediator (M)–outcome (Y) triad⁵⁵. In step 1 (MR for exposure (X) to mediator (M)), to select appropriate mediator candidates, we investigated the relationship between the genetically predicted LSA and modifiable risk factors. FDR correction was used to control for multiple comparisons of the IVW method results. In step 2 (MR for mediator (M) to exposure (X)), we conducted a bidirectional MR analysis to determine whether each significant risk factor in step 1 had an inverse effect on LSA. Risk factors without evidence of a potential causal effect on LSA were considered appropriate mediators. In step 3 (MR for mediator (M) to outcome (Y)), using the IVW method and FDR correction of 5% for mediators of each outcome, we estimated the effect of each mediator on each health outcome that was influenced by LSA in the primary analysis. In step 4, to assess the indirect effect of LSA on the pathway through the significant mediator–outcome pairs of step 3, we applied the ‘Product’ method that multiplies two coefficients obtained from step 1 and step 3 (known as ‘two-step MR’ or ‘network MR’ method)⁵⁵. We estimated the standard error of the indirect effect using the ‘Delta’ method⁵⁶. Finally, we estimated the proportion of the indirect effect of LSA, which was mediated by each risk factor, by dividing the indirect effect by the total effect.

Multivariable MR

The total effect estimates of the association between genetically predicted LSA and health outcomes can be mediated by other traits (unbalanced horizontal pleiotropy). Therefore, we undertook MVMR

analysis, a statistical method that allows for the association of IVs with multiple exposure traits to be combined into the analysis and examines direct effect estimates of each exposure on the outcome⁵⁷. Along with LSA (first exposure), each modifiable risk factor was included as another exposure (second exposure) to test whether LSA remain significant after adjusting for the effects of the modifiable risk factors. This analysis was conducted for exposure–mediator–outcome pairs that revealed significant associations in the primary analysis (MR for exposure (X) to outcome (Y)), step 1 (MR for exposure (X) to mediator (M)) and step 3 (MR for mediator (M) to outcome (Y)) of the mediation analysis. We harmonized the effects of each variant and the corresponding effect allele across the three datasets (two exposures and one outcome) and conducted additional LD clumping to identify IVs for MVMR.

Statistics and reproducibility

Statistical analysis was performed using R (<http://r-project.org>), PLINK version 1.90 (<https://www.cog-genomics.org/plink>), POLMM version 0.2.3 (<https://github.com/WenjianBI/POLMM>), LDSC version 1.0.1 (<https://github.com/bulik/ldsc>), FUMA (<https://fuma.ctglab.nl>), Two-SampleMR version 0.4.25 (<https://mrcieu.github.io/TwoSampleMR>) and MendelianRandomization version 0.9.0 (<https://cran.r-project.org/web/packages/MendelianRandomization/index.html>). Variant and sample QC in the UK Biobank was conducted following the UK Biobank QC documentation (<https://www.ukbiobank.ac.uk/wp-content/uploads/2014/04/>). FDR was used to adjust for multiple comparisons. No statistical method was used to pre-determine sample size. The investigators were not blinded to allocation during experiments and outcome assessment.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The UK Biobank data are publicly available upon reasonable application (<https://www.ukbiobank.ac.uk>). The GWAS summary statistics for this study were obtained from previously published studies (<https://www.ebi.ac.uk/gwas/summary-statistics>) and the MR-Base database (<https://www.mrbase.org>). The FUMA platform used the GTEx database v8 (<http://www.gtexportal.org/home/datasets>).

Code availability

Univariable and multivariable MR analyses were conducted using the R package ‘TwoSampleMR’ version 0.4.25 and ‘MendelianRandomization’ version 0.9.0, respectively. No custom codes were developed for this study. Forest plots were plotted using the R package ‘meta’ version 4.19.0.

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Author contributions

H.-H.W. and W.M. conceptualized and designed the study. S.K. conducted the statistical analysis. S.K. and H.K. wrote the first draft of the manuscript. All authors were involved in the interpretation of results and in reviewing and revising the manuscript and approved the final version. S.K. and H.K. are equal contributing authors, and H.-H.W. and W.M. are equal corresponding authors. The corresponding authors (H.-H.W. and W.M.) attest that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

Competing interests

All authors completed the International Committee of Medical Journal Editors uniform disclosure form at <https://www.icmje.org/disclosure-of-interest/> (available as Supplementary Table 19) and declare the following: W.-Y.P. is employed by a commercial company, GENINUS. P.N. reports investigator-initiated grants from Amgen, Apple, AstraZeneca, Boston Scientific and Novartis and personal fees from Apple, AstraZeneca, Blackstone Life Sciences, Foresite Labs, Novartis and Roche/Genentech. P.N. is also a co-founder of TenSixteen Bio and a scientific advisory board member of Esperion Therapeutics, geneXwell and TenSixteen Bio and reports spousal employment at Vertex, all unrelated to the present work. R.D. reports receiving grants from AstraZeneca and grants and non-financial support from Goldfinch Bio; being a scientific co-founder, consultant and equity holder at Pensieve Health; and being a consultant for Variant Bio, all unrelated to this work. All other authors declare no competing interests.

Additional information

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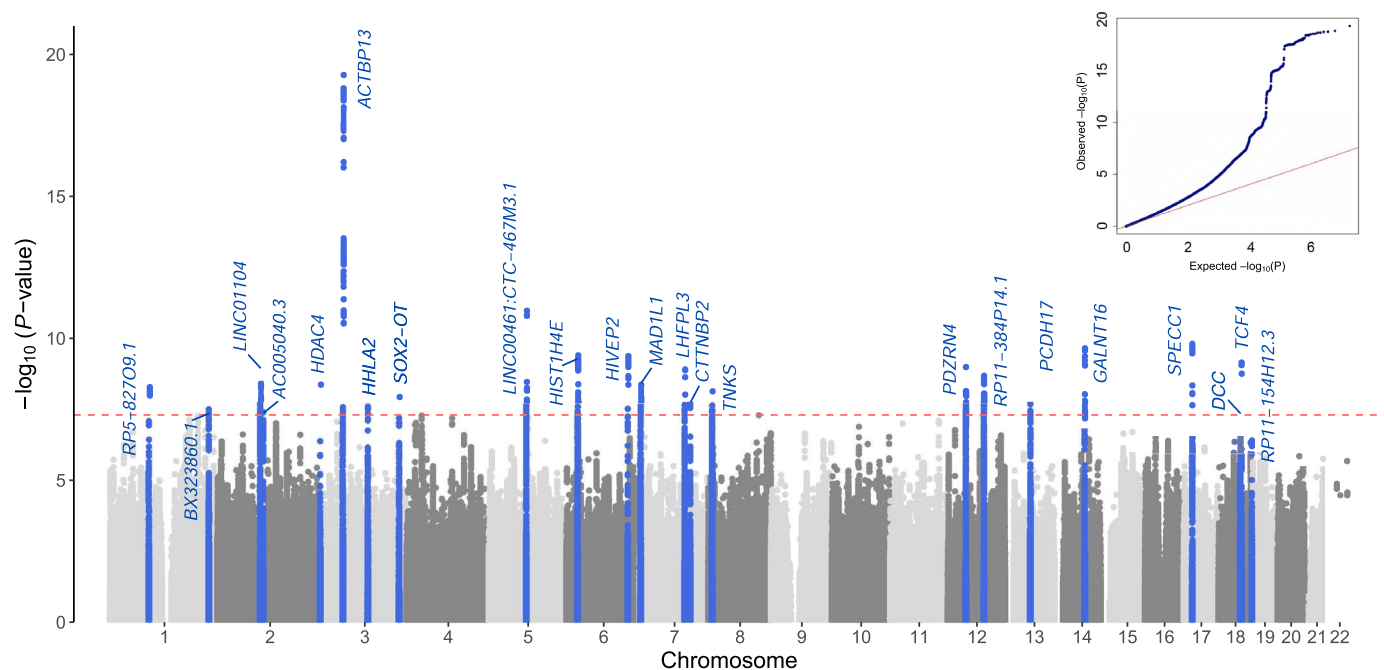
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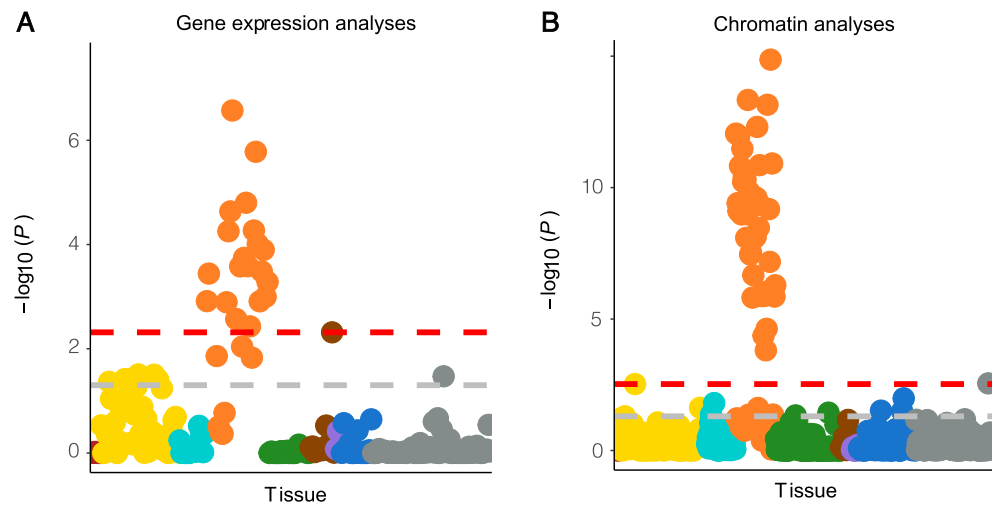
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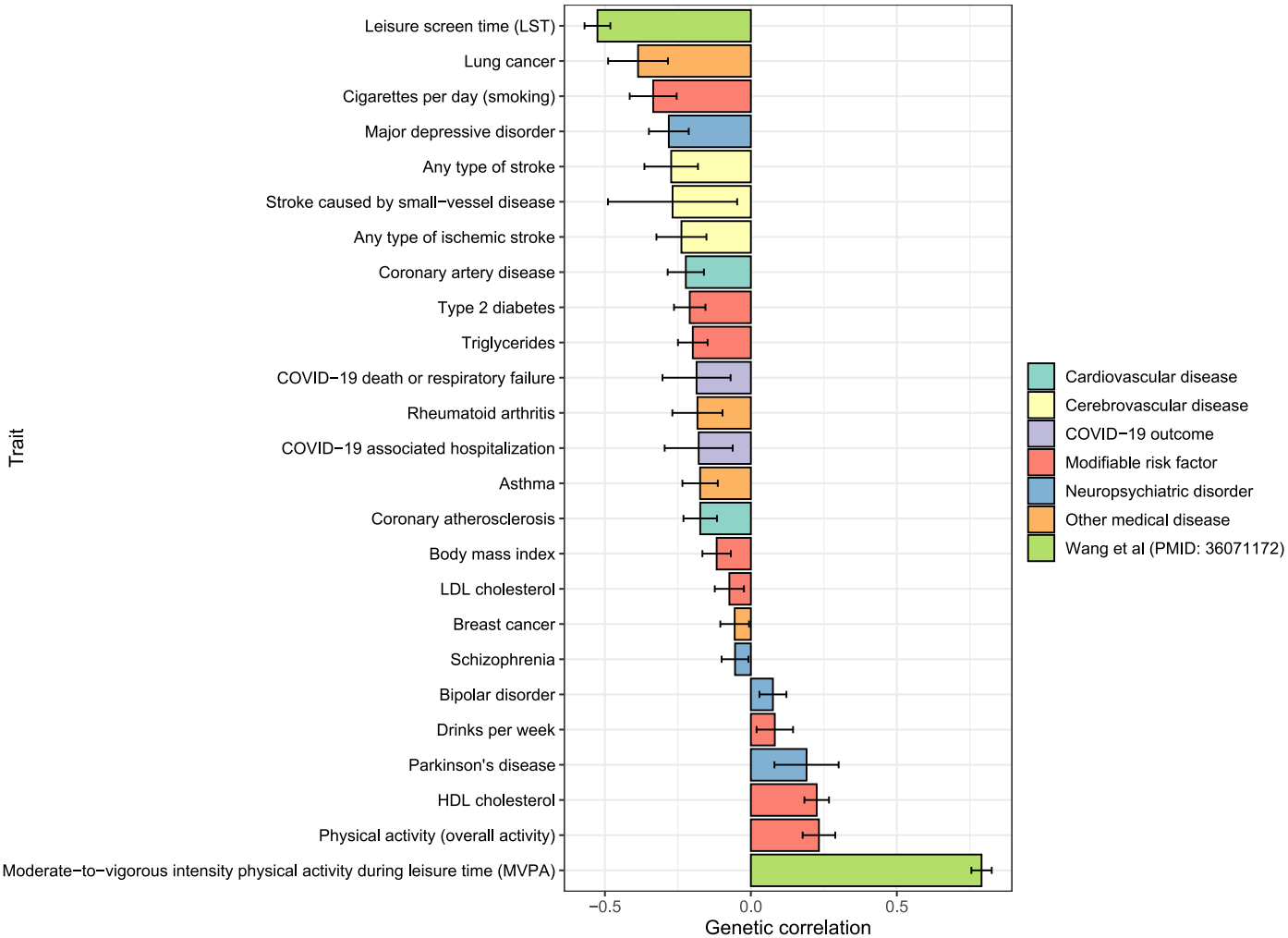
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Extended Data Fig. 1 | Manhattan plot and quantile-quantile (QQ) plot for LSA GWAS in the UK Biobank. Dashed horizontal line represents the genome-wide significant threshold ($P < 5 \times 10^{-8}$). Vertical and horizontal axes in the QQ plot mark the observed P values and the P values under the null hypothesis, respectively.

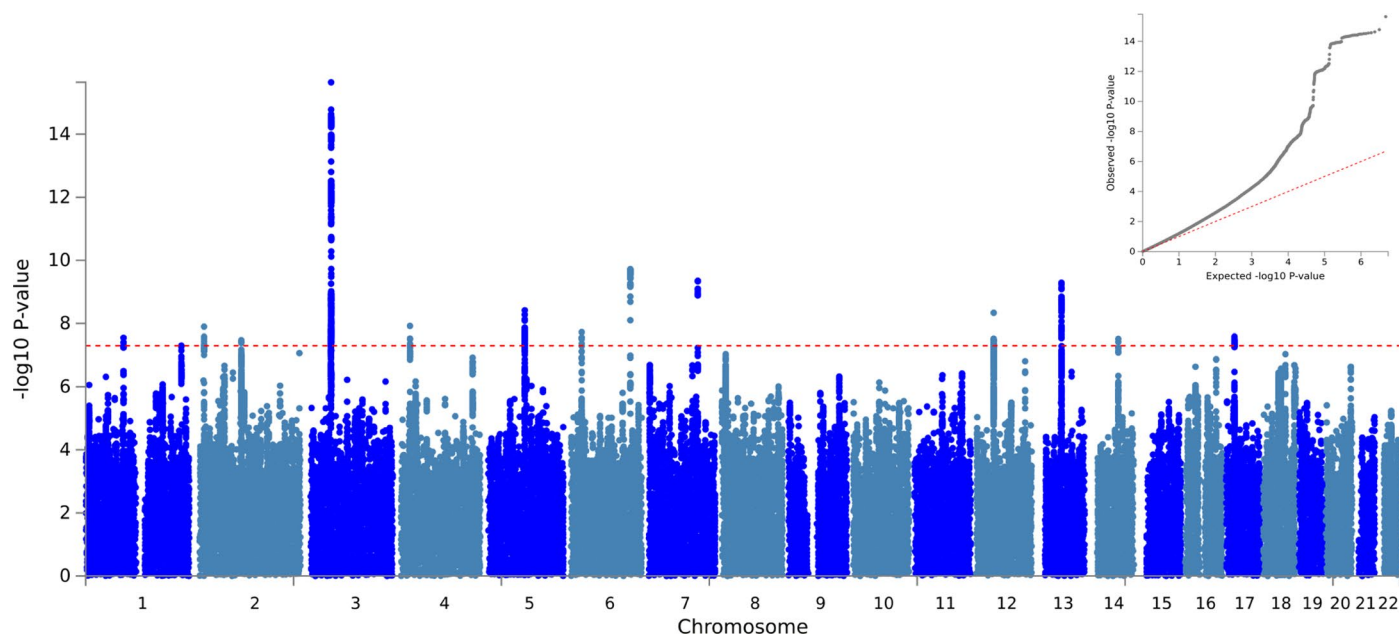


Extended Data Fig. 2 | Results of multiple-tissue analysis for LSA. Each circle represents a $-\log_{10} P$ -value for the one-sided LDSC-SEG enrichment test of gene expression (A) and chromatin analyses (B). The grey and red horizontal lines indicate a P -value of 0.05 and an FDR of 0.05, respectively.



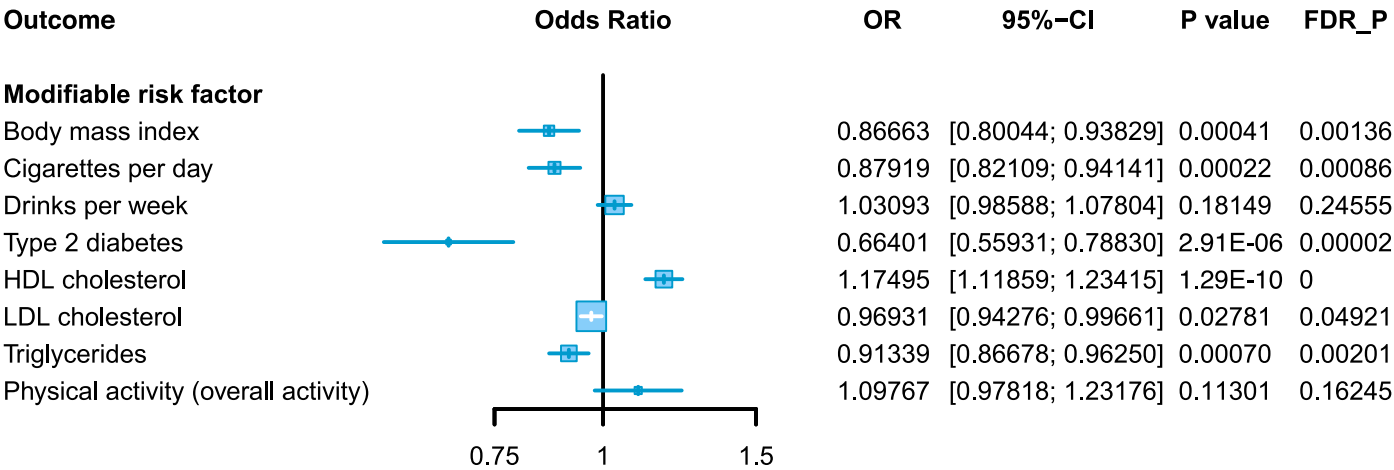
Extended Data Fig. 3 | Genetic correlations of traits significantly related to LSA at an FDR of <5%. Bars represent the genetic correlation estimates between LSA and each trait. A total of variants used in the analysis were $N_{\text{variant}} > 10$. Data are presented as mean $\pm$ standard deviation (s.d.). Statistical analyses were conducted using two-sided tests. Error bars represent the 95%

CIs of the estimated genetic correlations. This figure only includes significant genetic correlations (FDR <5%). Results for all traits included in the genetic correlation analysis are presented in Supplementary Table 7. COVID-19, Coronavirus disease 2019; LDL, low-density lipoprotein; HDL, high-density lipoprotein



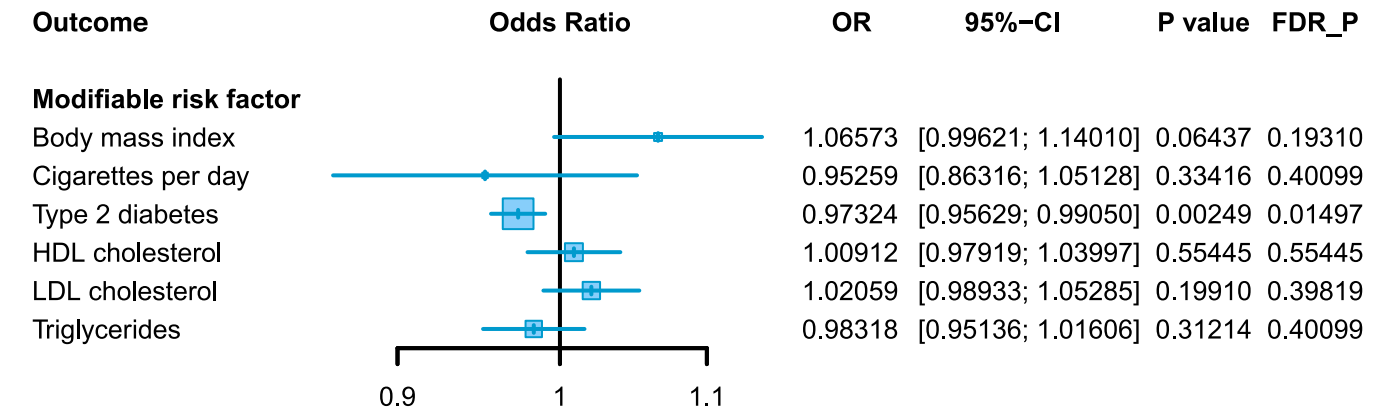
Extended Data Fig. 4 | Manhattan plot and quantile-quantile (QQ) plot for LSA GWAS in the UK Biobank after removing the sample overlap between LSA data and physical activity data ($n = 86,872$). The total sample size used in the GWAS was 320,262. Dashed horizontal line represents the genome-

wide significant threshold ($P < 5 \times 10^{-8}$). Vertical and horizontal axes in the QQ plot mark the observed P values and the P values under the null hypothesis, respectively.

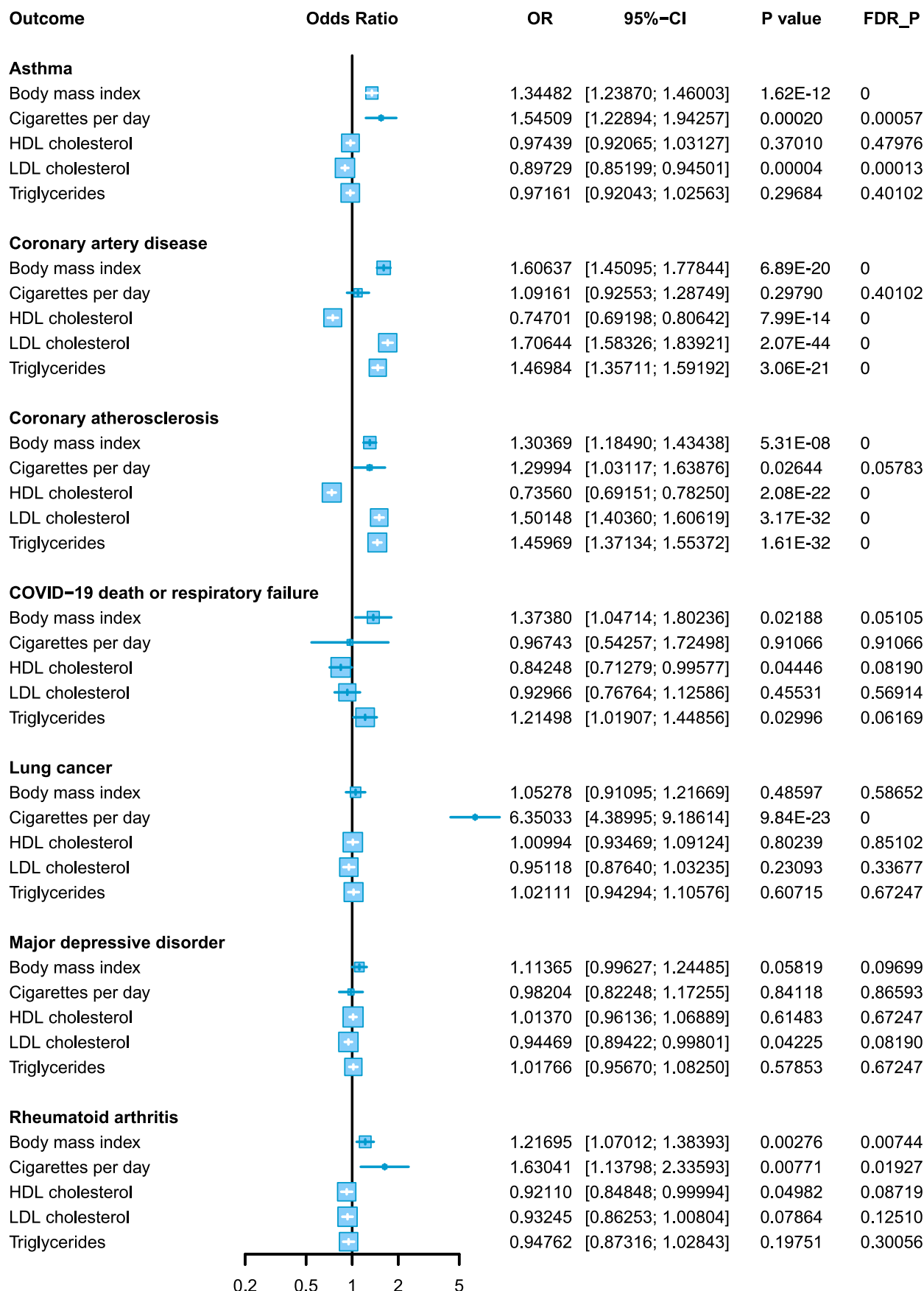


Extended Data Fig. 5 | Forest plot for the univariable Mendelian randomisation showing the effects of genetically predicted LSA on modifiable risk factors. The risk factors included in the analyses were limited to traits genetically correlated with LSA. A total of variants used in the analysis were

$N_{\text{variant}} > 10$. Data are presented as mean $\pm$ standard deviation (s.d.). Statistical analyses were conducted using two-sided tests. Error bars represent the 95% CIs of the effects. OR, odds ratio; CI, confidence interval; P, *P*-value; FDR_P, false discovery rate (FDR)-corrected *P*-value.



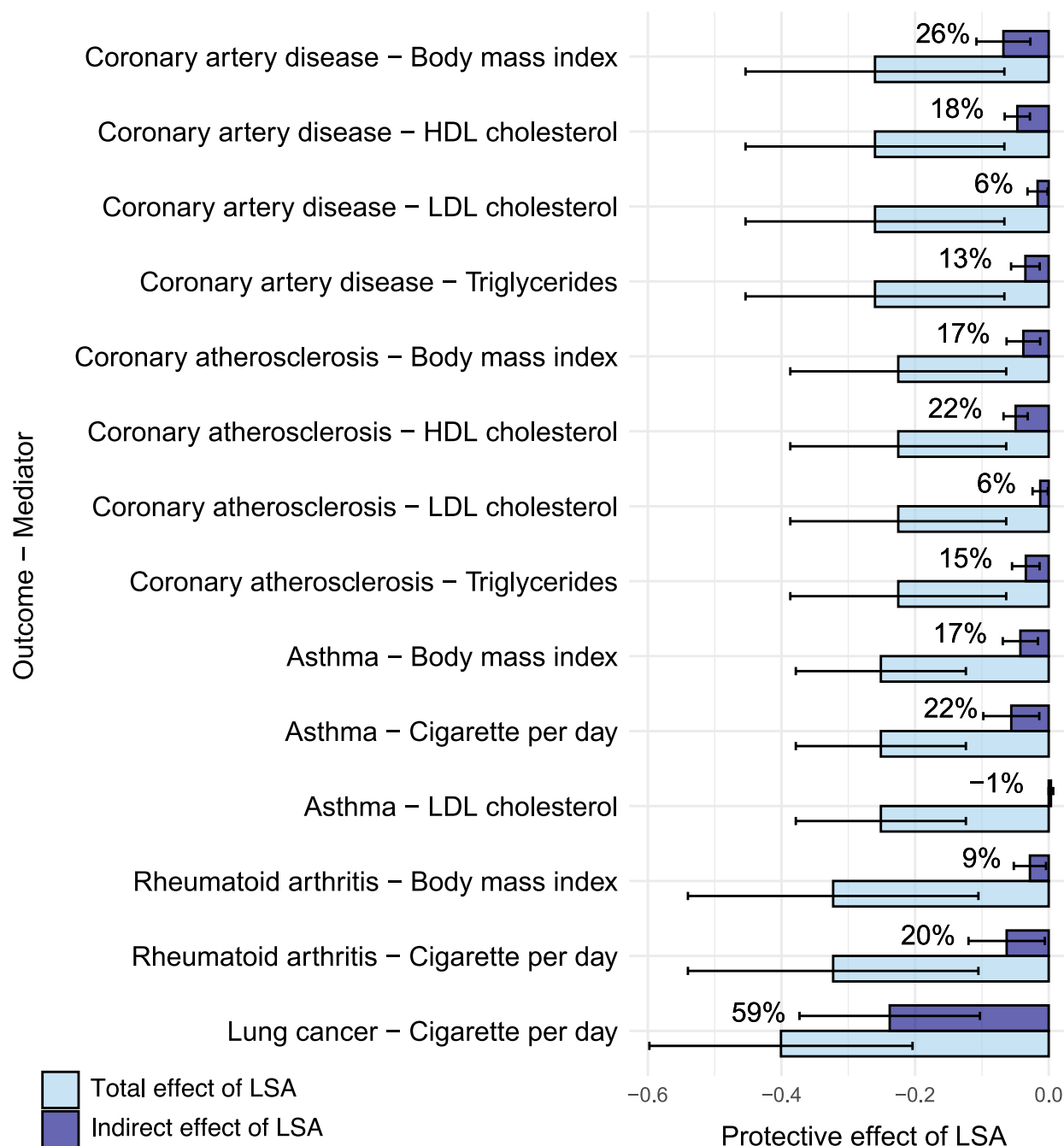
Extended Data Fig. 6 | Forest plot for the univariable Mendelian randomisation showing the impact of genetically predicted risk factors on LSA. The risk factors included in the analyses were limited to traits substantially affected by LSA. A total of variants used in the analysis were $N_{\text{variant}} > 10$. Data are presented as mean $\pm$ standard deviation (s.d.). Statistical analyses were conducted using two-sided tests. Error bars represent the 95% CIs of the effects. OR, odds ratio; CI, confidence interval; P, *P*-value; FDR_P, false discovery rate (FDR)-corrected *P*-value.



Extended Data Fig. 7 | See next page for caption.

Extended Data Fig. 7 | Forest plot for the univariable Mendelian randomisation showing the effects of genetically predicted risk factors on health outcomes. In the analyses, we included five risk factors and seven health outcomes that were significantly affected by LSA (FDR-corrected $P < 0.05$). A total of variants used in the analysis were $N_{\text{variant}} > 10$. Data are

presented as mean $\pm$ standard deviation (s.d.). Statistical analyses were conducted using two-sided tests. Error bars represent the 95% CIs of the effects. OR, odds ratio; CI, confidence interval; P, P -value; FDR_P, false discovery rate (FDR)-corrected P -value; COVID-19, Coronavirus disease 2019.



Extended Data Fig. 8 | Total and indirect effects of LSA on five health outcomes via selected mediators. The percentage on the bar indicates the estimated proportion of the indirect effect via a selected mediator. A total

of variants used in the analysis were $N_{\text{variant}} > 10$. Data are presented as mean $\pm$ standard deviation (s.d.). Statistical analyses were conducted using two-sided tests. Error bars represent the 95% CIs of the effects.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	This research has been conducted using the UK Biobank Resource (application number 33022) and previously reported genome-wide association study (GWAS) summary statistics (sample size = 27,209-1,557,411) that did not include the UK Biobank data. No software was used for data collection.
Data analysis	SHAPEIT3, https://jmarchini.org/software/#shapeit-3 ; IMPUTE2, https://jmarchini.org/software/#impute-2 ; PLINK v1.90, https://www.cog-genomics.org/plink ; POLMM v0.2.3, https://github.com/WenjianBI/POLMM ; LDSC v1.0.1, https://github.com/bulik/ldsc ; FUMA, https://fuma.ctglab.nl ; MR base, https://www.mrbase.org ; TwoSampleMR v0.4.25, https://mrcieu.github.io/TwoSampleMR ; MendelianRandomization v0.9.0, https://cran.r-project.org/web/packages/MendelianRandomization/index.html ; MR-PRESSO v1, https://github.com/rondolab/MR-PRESSO

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The UK Biobank data are publicly available upon reasonable application (<https://www.ukbiobank.ac.uk>). The GWAS summary statistics for this study were obtained from previously published studies (<https://www.ebi.ac.uk/gwas/summary-statistics>) and the MR-Base database (<https://www.mrbase.org>). The FUMA platform utilized the GTEx database v8 (<http://www.gtexportal.org/home/datasets>).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	We used sex as a covariate to adjust our analyses. In total, 187,065 males (45.9%) and 220,069 (54.1%) females were included in the European GWAS. There is no mention of gender in this study.
Reporting on race, ethnicity, or other socially relevant groupings	Among the 487,409 individuals in the UK Biobank, a GWAS was conducted for 407,134 individuals of European ancestry.
Population characteristics	Analyses used age, sex, and principal components from the genetic data as covariates to control for the population structure. The total sample size was 407,134 in the UK Biobank, consisting of 187,065 males (45.9%) and 220,069 (54.1%) females.
Recruitment	Researchers in this study were not involved in the participant recruitment.
Ethics oversight	The study was approved by the Institutional Review Board of Seoul National University Bundang Hospital (X-2202-737-902). Each GWAS summary statistic used in this study was previously approved by its respective research ethics committee (see Table 1). The UK Biobank study was approved by the National Research Ethics Committee (REC reference 11/NW/0382). All UK Biobank participants provided informed consent. The UK Biobank data were obtained under application number 33002.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Describe how sample size was determined, detailing any statistical methods used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.
Data exclusions	Describe any data exclusions. If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.
Replication	Describe the measures taken to verify the reproducibility of the experimental findings. If all attempts at replication were successful, confirm this OR if there are any findings that were not replicated or cannot be reproduced, note this and describe why.
Randomization	Describe how samples/organisms/participants were allocated into experimental groups. If allocation was not random, describe how covariates were controlled OR if this is not relevant to your study, explain why.
Blinding	Describe whether the investigators were blinded to group allocation during data collection and/or analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This is a GWAS for leisure and social activities in European population, followed by Mendelian randomization analyses. The data used in this study are quantitative.
Research sample	The total sample size was 407,134 in the UK Biobank, consisting of 187,065 males (45.9%) and 220,069 (54.1%) females.
Sampling strategy	We did not apply a separate sampling procedure, as our data source is the UK Biobank. All individual data was used after applying exclusion criteria described in the manuscript. The total sample size of the current study was the largest sample size of non-overlapping samples used to date, and gave sufficient statistics for the effect sizes of associations.
Data collection	The UK Biobank is a nation-wide prospective study-based database involving >500,000 individuals between ages of 40–69 years recruited in multiple centres throughout the UK. The UK Biobank has collected electronic medical records, computer-assisted interview data, touchscreen-based self-reported questionnaire data, physical and functional measures, and biological samples, including genotype data. No additional individuals were present besides the participants and the researchers, and the researchers were not blinded to the study hypothesis.
Timing	The UK Biobank data were collected from 2006 to 2010.
Data exclusions	Participants with missing data or ‘prefer not to answer’ were excluded, leaving 407,134 for analysis. Detailed information is described in the Methods section.
Non-participation	No participants dropped out or declined participation.
Randomization	Randomization is not relevant to this study because we utilized GWAS summary statistics. Since summary statistics are derived from previously conducted genome-wide association studies, they represent aggregated data rather than individual-level data. This approach focuses on identifying associations between genetic variants and traits, which does not require randomization.

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	<i>Briefly describe the study. For quantitative data include treatment factors and interactions, design structure (e.g. factorial, nested, hierarchical), nature and number of experimental units and replicates.</i>
Research sample	<i>Describe the research sample (e.g. a group of tagged <i>Passer domesticus</i>, all <i>Stenocereus thurberi</i> within Organ Pipe Cactus National Monument), and provide a rationale for the sample choice. When relevant, describe the organism taxa, source, sex, age range and any manipulations. State what population the sample is meant to represent when applicable. For studies involving existing datasets, describe the data and its source.</i>
Sampling strategy	<i>Note the sampling procedure. Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.</i>
Data collection	<i>Describe the data collection procedure, including who recorded the data and how.</i>
Timing and spatial scale	<i>Indicate the start and stop dates of data collection, noting the frequency and periodicity of sampling and providing a rationale for these choices. If there is a gap between collection periods, state the dates for each sample cohort. Specify the spatial scale from which the data are taken</i>
Data exclusions	<i>If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.</i>
Reproducibility	<i>Describe the measures taken to verify the reproducibility of experimental findings. For each experiment, note whether any attempts to repeat the experiment failed OR state that all attempts to repeat the experiment were successful.</i>
Randomization	<i>Describe how samples/organisms/participants were allocated into groups. If allocation was not random, describe how covariates were controlled. If this is not relevant to your study, explain why.</i>
Blinding	<i>Describe the extent of blinding used during data acquisition and analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.</i>
Did the study involve field work?	☐ Yes ☐ No

Field work, collection and transport

Field conditions	<i>Describe the study conditions for field work, providing relevant parameters (e.g. temperature, rainfall).</i>
Location	<i>State the location of the sampling or experiment, providing relevant parameters (e.g. latitude and longitude, elevation, water depth).</i>
Access & import/export	<i>Describe the efforts you have made to access habitats and to collect and import/export your samples in a responsible manner and in compliance with local, national and international laws, noting any permits that were obtained (give the name of the issuing authority, the date of issue, and any identifying information).</i>
Disturbance	<i>Describe any disturbance caused by the study and how it was minimized.</i>

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	<i>Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.</i>
Validation	<i>Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.</i>

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	<i>State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.</i>
Authentication	<i>Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.</i>
Mycoplasma contamination	<i>Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.</i>
Commonly misidentified lines (See ICLAC register)	<i>Name any commonly misidentified cell lines used in the study and provide a rationale for their use.</i>

Palaeontology and Archaeology

Specimen provenance	<i>Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.</i>
Specimen deposition	<i>Indicate where the specimens have been deposited to permit free access by other researchers.</i>

Dating methods

If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight

Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals

For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.

Wild animals

Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.

Reporting on sex

Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.

Field-collected samples

For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.

Ethics oversight

Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration

Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.

Study protocol

Note where the full trial protocol can be accessed OR if not available, explain why.

Data collection

Describe the settings and locales of data collection, noting the time periods of recruitment and data collection.

Outcomes

Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.

Dual use research of concern

Policy information about [dual use research of concern](#)

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes
☒	☐ Public health
☒	☐ National security
☒	☐ Crops and/or livestock
☒	☐ Ecosystems
☒	☐ Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes
☒	☐ Demonstrate how to render a vaccine ineffective
☒	☐ Confer resistance to therapeutically useful antibiotics or antiviral agents
☒	☐ Enhance the virulence of a pathogen or render a nonpathogen virulent
☒	☐ Increase transmissibility of a pathogen
☒	☐ Alter the host range of a pathogen
☒	☐ Enable evasion of diagnostic/detection modalities
☒	☐ Enable the weaponization of a biological agent or toxin
☒	☐ Any other potentially harmful combination of experiments and agents

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.

ChIP-seq

Data deposition

- ☐ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links <i>May remain private before publication.</i>	For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.
Files in database submission	Provide a list of all files available in the database submission.
Genome browser session (e.g. UCSC)	Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.

Methodology

Replicates	Describe the experimental replicates, specifying number, type and replicate agreement.
Sequencing depth	Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.
Antibodies	Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.
Peak calling parameters	Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.
Data quality	Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.
Software	Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.

Flow Cytometry

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.

Instrument

Identify the instrument used for data collection, specifying make and model number.

Software

Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.

Cell population abundance

Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.

Gating strategy

Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.

- ☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type

Indicate task or resting state; event-related or block design.

Design specifications

Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.

Behavioral performance measures

State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).

Acquisition

Imaging type(s)

Specify: functional, structural, diffusion, perfusion.

Field strength

Specify in Tesla

Sequence & imaging parameters

Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.

Area of acquisition

State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.

Diffusion MRI

☐ Used

☐ Not used

Preprocessing

Preprocessing software

Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).

Normalization

If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.

Normalization template

Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.

Noise and artifact removal

Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).

Volume censoring

Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.

Statistical modeling & inference

Model type and settings

Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).

Effect(s) tested

Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.

(See [Eklund et al. 2016](#))

Correction

Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).

Models & analysis

n/a | Involved in the study

☐☐ Functional and/or effective connectivity☐☐ Graph analysis☐☐ Multivariate modeling or predictive analysis

Functional and/or effective connectivity

Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).

Graph analysis

Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).

Multivariate modeling and predictive analysis

Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.